

CSE 311:

Data Communication

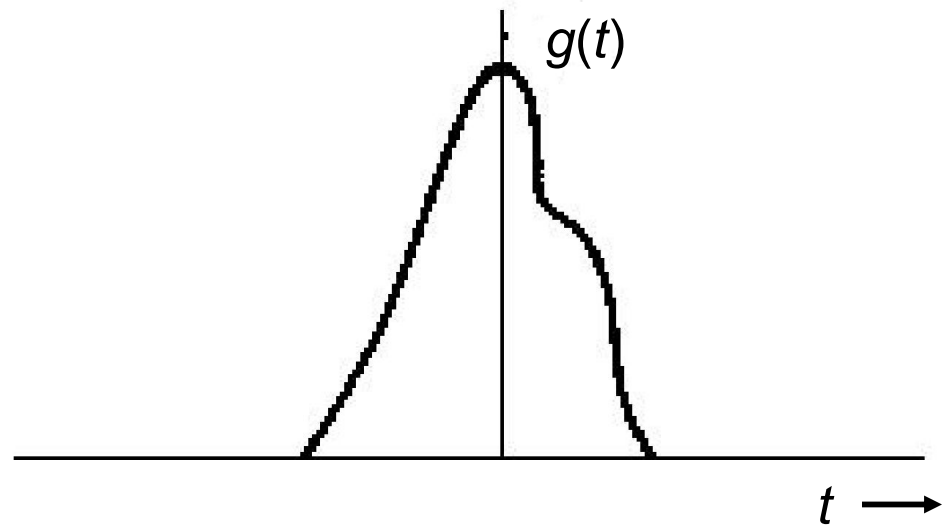
Instructor:
Dr. Md. Monirul Islam

Signals and systems

Topics

- Definition, classification and properties of signals
- Examples of some useful signals
- Fourier Series of Periodic Signals

Fourier's theorem for aperiodic Signal



Fourier's theorem for aperiodic Signal

- *aperiodic* signal but *with finite area under curve* can be represented as *integral* of sines and cosines

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j\omega t} df$$

$$\text{where, } G(f) = \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt$$

$$\text{and } \omega = 2\pi f$$

Fourier's theorem for aperiodic Signal

- These two are called Fourier transform pair

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} g(t) e^{-j2\pi ft} dt$$
$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j\omega t} df = \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df$$

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Symbolically shown as,

$$g(t) \Leftrightarrow G(f) \text{ or } G(f) = \mathfrak{F}[g(t)]$$

Fourier Integral for aperiodic signal

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-j2\pi ft} dt$$

- $G(f)$ is Fourier spectra and complex even if $g(t)$ is real
- $G(f)$ can be shown as

$$G(f) = |G(f)| e^{j\theta_g(f)}$$

Fourier Integral for aperiodic signal

$$G(f) = \int_{-\infty}^{\infty} g(t)e^{-j\omega t} dt = \int_{-\infty}^{\infty} g(t)e^{-j2\pi ft} dt$$

- Replacing f by $-f$

$$G(-f) = \int_{-\infty}^{\infty} g(t)e^{j2\pi ft} dt$$

Fourier Integral for aperiodic signal

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$$G(-f) = \int_{-\infty}^{\infty} g(t)e^{j2\pi ft} dt$$

$$G^*(f) = \int_{-\infty}^{\infty} g(t)e^{j2\pi ft} dt \quad \text{provided, } g(t) \text{ is real}$$

Fourier Integral for aperiodic signal

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- Replacing f by $-f$

$$G(-f) = \int_{-\infty}^{\infty} g(t)e^{j2\pi ft} dt$$

$$G^*(f) = \int_{-\infty}^{\infty} g(t)e^{j2\pi ft} dt = G(-f)$$

Fourier Integral for aperiodic signal

- The following is complex conjugate symmetry

$$G^*(f) = G(-f)$$

Fourier Integral for aperiodic signal

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Therefore,

$$|G(f)| = |G^*(f)| = |G(-f)|$$

Fourier Integral for aperiodic signal

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$$G^*(f) = G(-f)$$

Therefore,

$$|G(f)| = |G^*(f)| = |G(-f)|$$

but

$$\theta_g(-f) = -\theta_g(f)$$

Conjugate Symmetry Property in Fourier Series of Periodic signals

Revisited

- D_n are complex though $\phi(t)$ is a **real** periodic
 - $D_n = |D_n|e^{j\angle D_n}$
- D_n and D_{-n} are complex conjugates
 - $|D_n| = |D_{-n}|$
 - $\angle D_n = \theta_n$ and $\angle D_{-n} = -\theta_n$
- Therefore, $D_n = |D_n|e^{j\theta_n}$ and $D_{-n} = |D_n|e^{-j\theta_n}$

Conjugate Symmetry Property in Fourier Series of Periodic signals

Revisited!

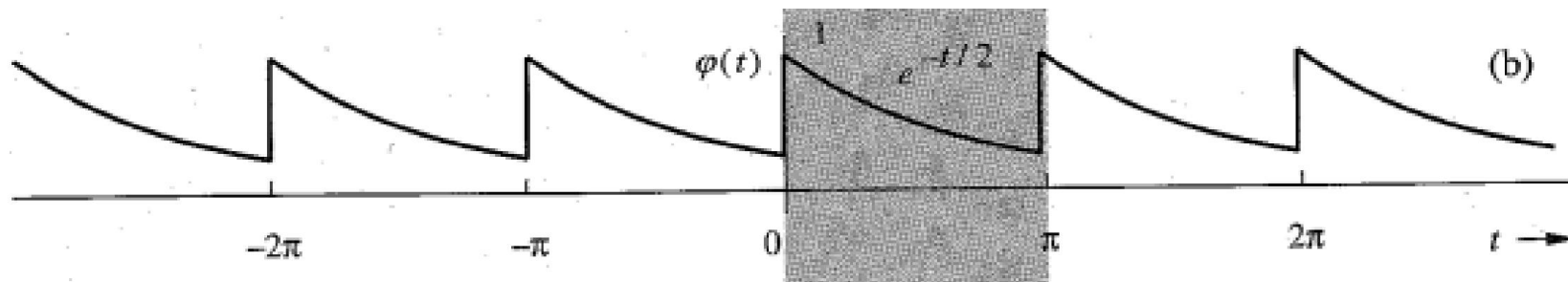
$$|D_n| = |D_{-n}|$$

$$\angle D_n = \theta_n \quad \text{and} \quad \angle D_{-n} = -\theta_n$$

- $|D_n|$ are amplitudes and $\angle D_n$ are angles of exponential spectra
- $|D_n|$ vs. f is an even function
- $\angle D_n$ vs. f is an odd function

Conjugate Symmetry Property in Fourier Series of Periodic signals

Revisited!

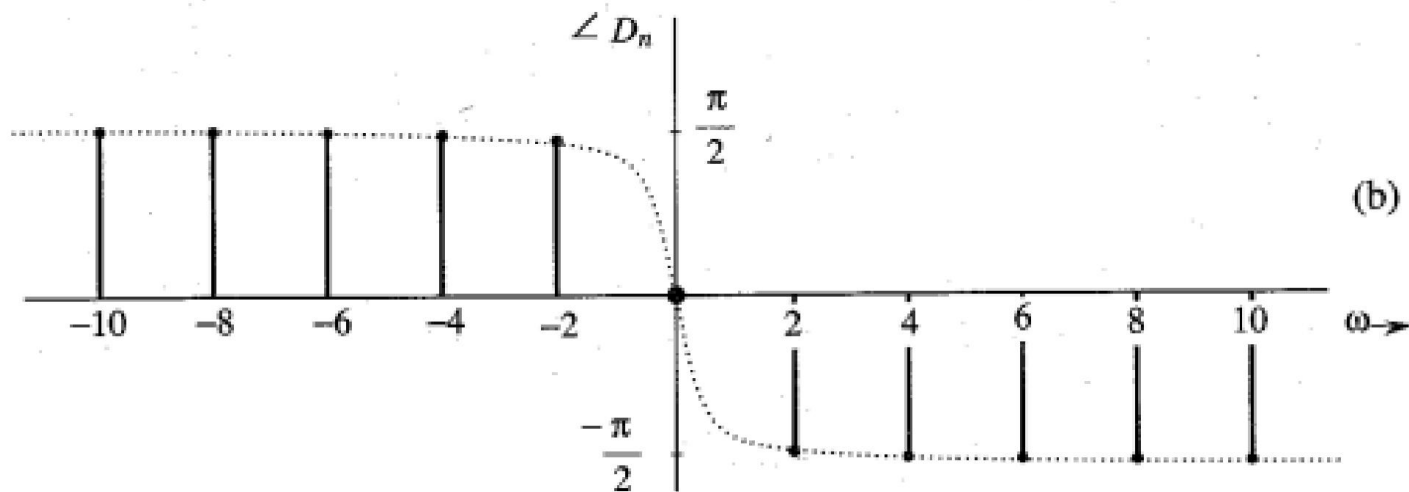
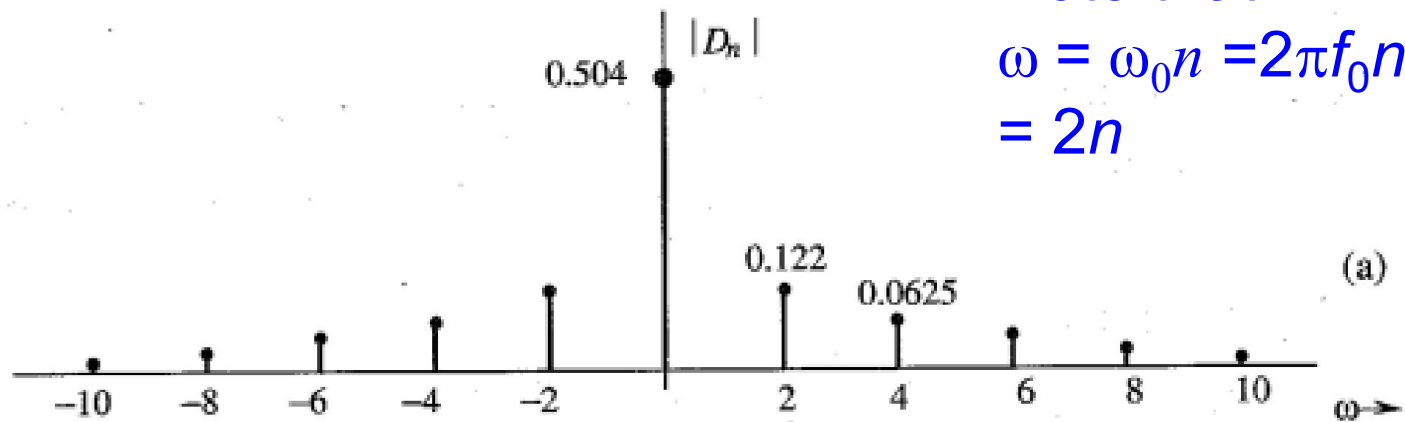


$$\phi(t) = \sum_{n=-\infty}^{n=\infty} D_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{n=\infty} D_n e^{j2nt}$$

Conjugate Symmetry Property in Fourier Series of Periodic signals

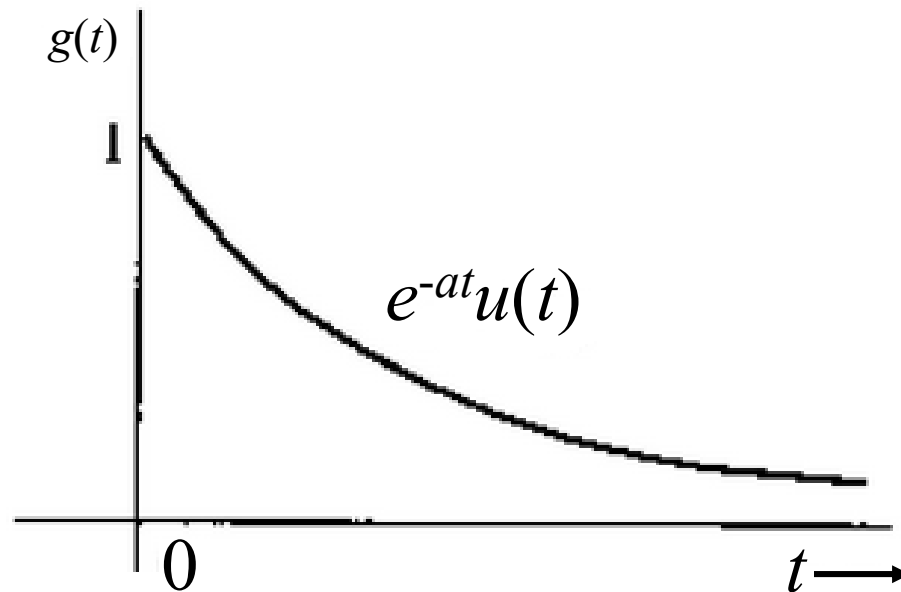
Revisited!

Note that
 $\omega = \omega_0 n = 2\pi f_0 n$
 $= 2n$



Fourier Transform Example

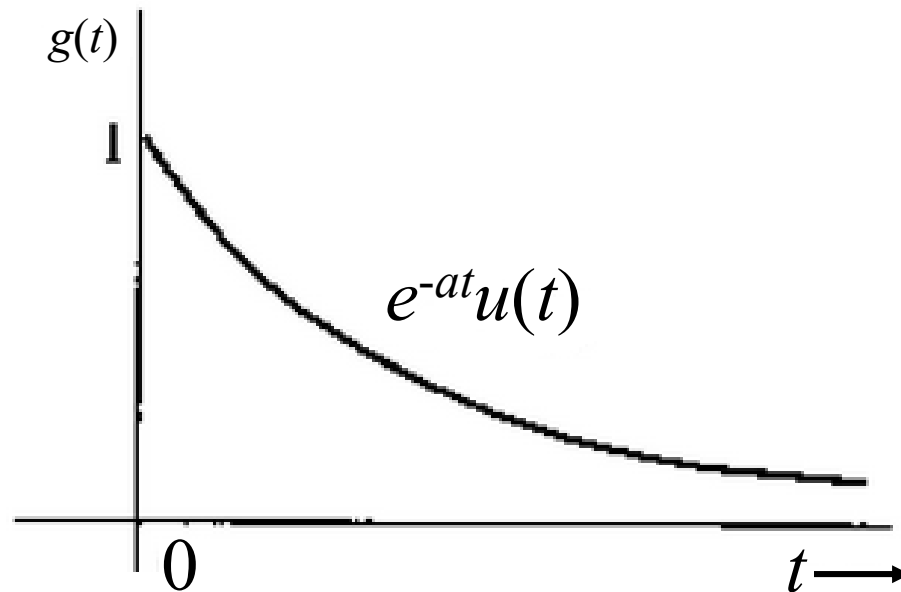
Find Fourier transform of $e^{-at}u(t)$



Fourier Transform Example

Find Fourier transform of $e^{-at}u(t)$

$$G(f) = \int_{-\infty}^{\infty} g(t)e^{-j2\pi ft} dt = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-j2\pi ft} dt = \int_0^{\infty} e^{-at}e^{-j2\pi ft} dt$$

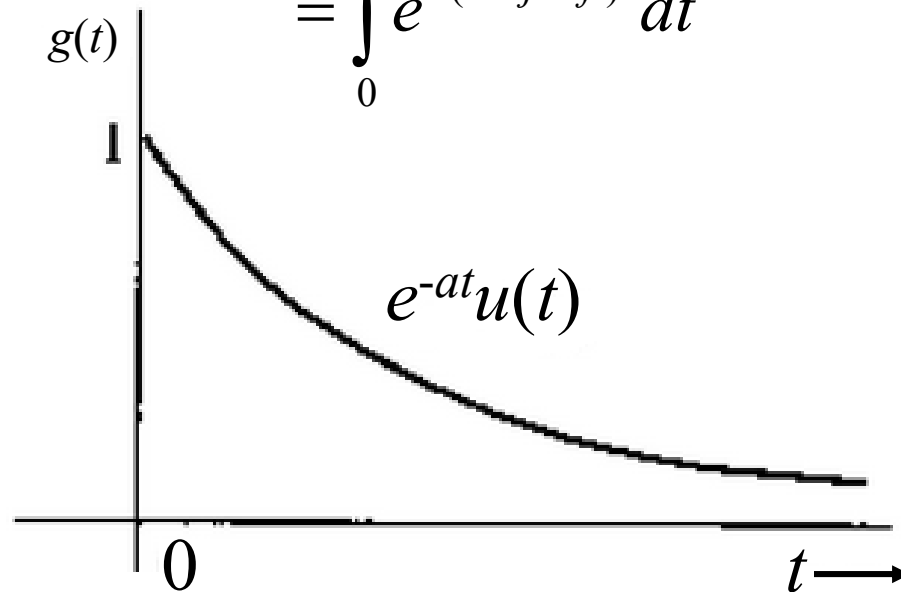


Fourier Transform Example

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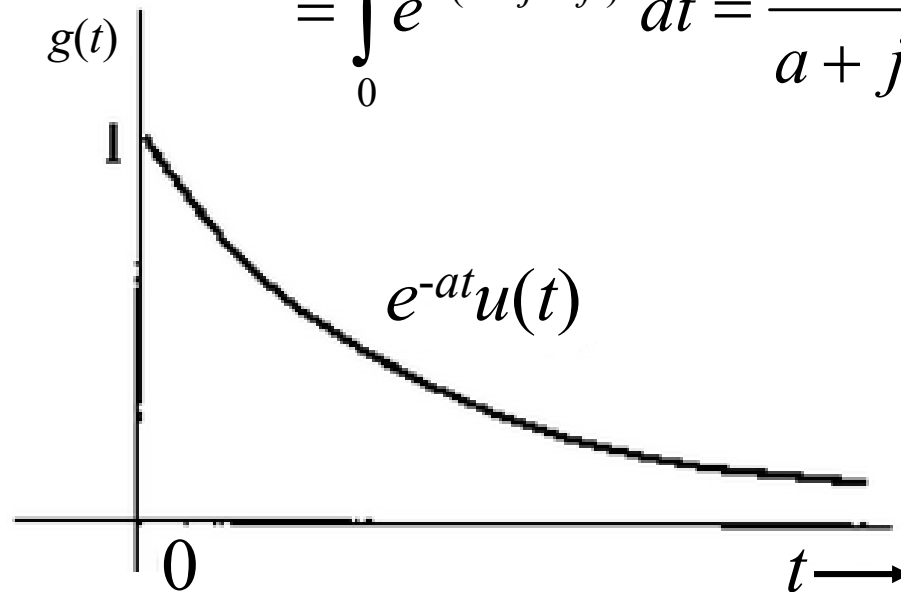
$$= \int_0^{\infty} e^{-(a+j2\pi f)t} dt$$



Fourier Transform Example

Find Fourier transform of $e^{-at}u(t)$

$$\begin{aligned} G(f) &= \int_{-\infty}^{\infty} g(t) e^{-j2\pi f t} dt = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-j2\pi f t} dt = \int_0^{\infty} e^{-at} e^{-j2\pi f t} dt \\ &= \int_0^{\infty} e^{-(a+j2\pi f)t} dt = \frac{-1}{a+j2\pi f} \left[e^{-(a+j2\pi f)t} \right]_0^{\infty} \end{aligned}$$



Fourier Transform Example

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Now, as $t \rightarrow \infty$, $e^{-(a+j2\pi f)t} = e^{-at} e^{-j2\pi f t} = 0$

Fourier Transform Example

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Now, as $t \rightarrow \infty$, $e^{-(a+j2\pi f)t} = e^{-at} e^{-j2\pi f t} = 0$ provided $a > 0$

Fourier Transform Example

Find Fourier transform of $e^{-at}u(t)$

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Now, as $t \rightarrow \infty$, $e^{-(a+j2\pi f)t} = e^{-at} e^{-j2\pi ft} = 0$

Fourier Transform Example

Find Fourier transform of $e^{-at}u(t)$

$$G(f) = \frac{1}{a + j\omega} = \frac{1}{a + j2\pi f}$$

$$G(f) = \frac{1}{\sqrt{a^2 + (2\pi f)^2}} e^{-j \tan^{-1} \left(\frac{2\pi f}{a} \right)}$$

Fourier Transform Example

Find Fourier transform of $e^{-at}u(t)$

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Therefore,

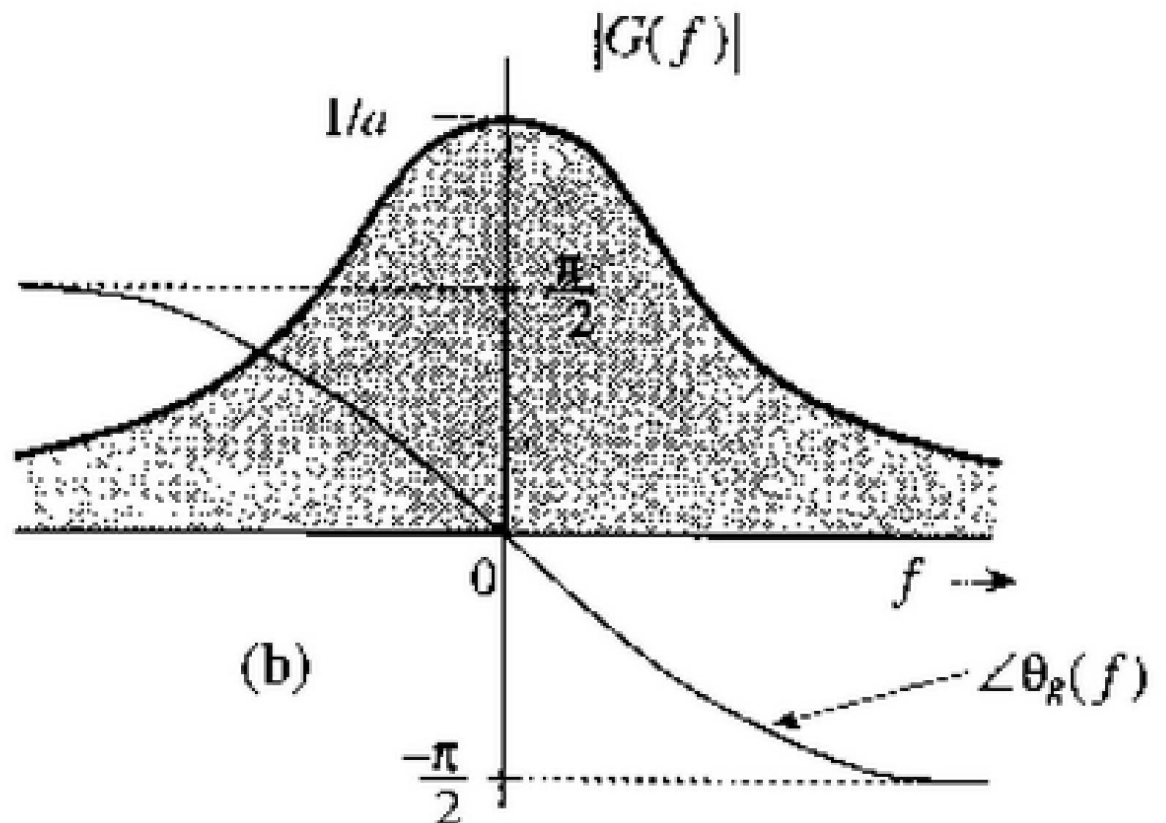
$$|G(f)| = \frac{1}{\sqrt{a^2 + (2\pi f)^2}} \quad \text{and} \quad \theta_g(f) = -\tan^{-1} \left(\frac{2\pi f}{a} \right)$$

Fourier Transform Example

Find Fourier transform of $e^{-at}u(t)$

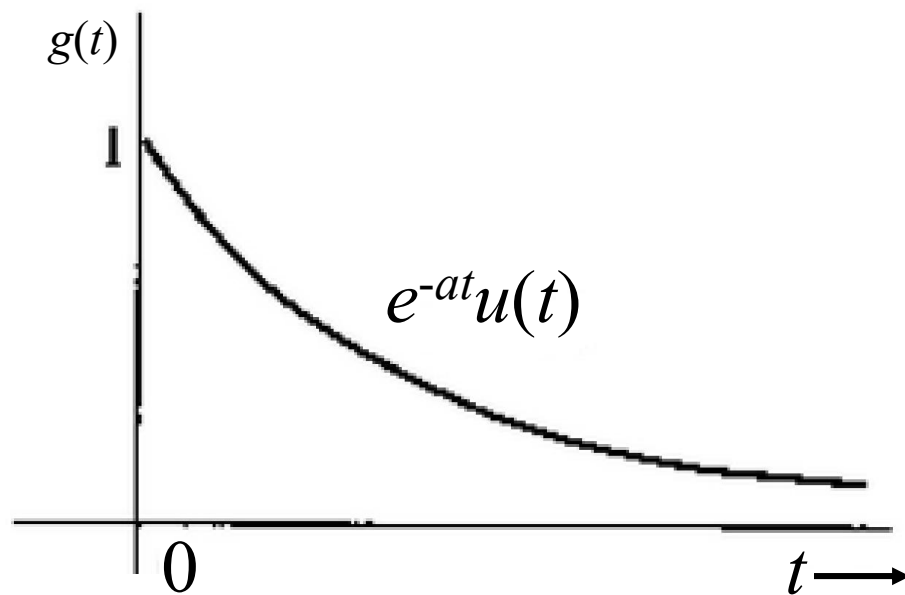
$$|G(f)| = \frac{1}{\sqrt{a^2 + (2\pi f)^2}}$$

$$\theta_g(f) = -\tan^{-1}\left(\frac{2\pi f}{a}\right)$$



Fourier Transform Example

Remember, we got this result for $a > 0$

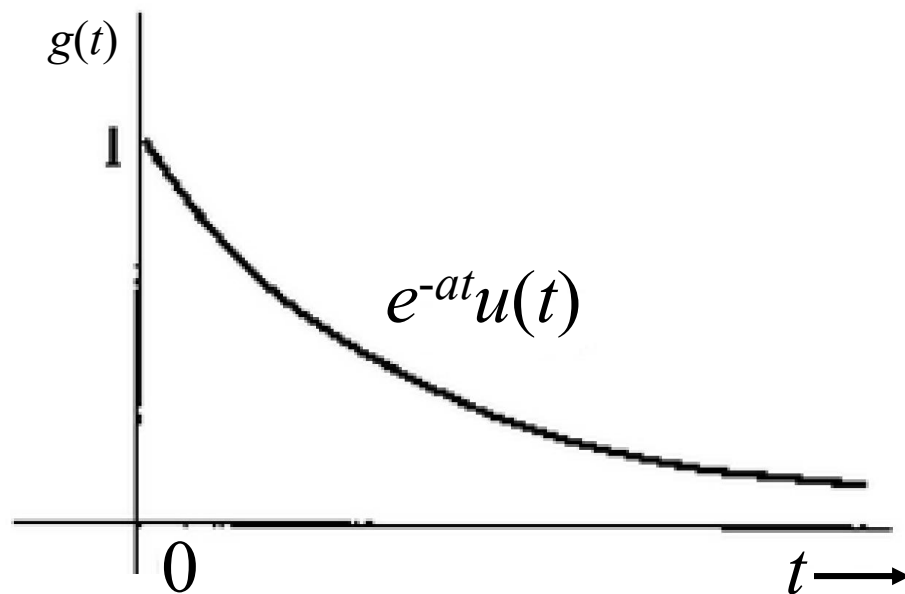


$$\begin{aligned} G(f) &= \frac{-1}{a + j2\pi f} \left[e^{-(a+j2\pi f)t} \right]_0^\infty \\ &= \frac{1}{a + j2\pi f} \end{aligned}$$

Now as $t \rightarrow \infty$, $e^{-(a+j2\pi f)t} = e^{-at} e^{-j2\pi f t} = 0$ provided $a > 0$

Fourier Transform Example

What happens if result for $a < 0$?



$$\begin{aligned} G(f) &= \frac{-1}{a + j2\pi f} \left[e^{-(a+j2\pi f)t} \right]_0^\infty \\ &= \frac{1}{a + j2\pi f} \end{aligned}$$

Now as $t \rightarrow \infty$, $e^{-(a+j2\pi f)t} = e^{-at} e^{-j2\pi f t} = 0$ provided $a > 0$

Existence of Fourier Transform

Conditions

- Area under curve must be finite
- The function must be integrable

$$\int_{-\infty}^{\infty} |g(t)| dt < \infty$$

Dirichlet condition

Existence of Fourier Transform

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$$\int_{-\infty}^{\infty} |g(t)| dt < \infty$$

This can be proved from,

$$|G(f)| = \left| \int_{-\infty}^{\infty} g(t) e^{-j2\pi ft} dt \right|$$

Existence of Fourier Transform

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$$|G(f)| = \left| \int_{-\infty}^{\infty} g(t) e^{-j2\pi ft} dt \right| \leq \int_{-\infty}^{\infty} |g(t)| \times |e^{-j2\pi ft}| dt$$

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$$|G(f)| \leq \int_{-\infty}^{\infty} |g(t)| dt$$

Linearity of Fourier Transform

Assume, $g_1(t) \Leftrightarrow G_1(f)$ and $g_2(t) \Leftrightarrow G_2(f)$

and a_1 and a_2 are arbitrary constants

Then,

$$a_1 g_1(t) + a_2 g_2(t) \Leftrightarrow a_1 G_1(f) + a_2 G_2(f)$$

Linearity of Fourier Transform

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Then,

$$a_1 g_1(t) + a_2 g_2(t) \Leftrightarrow a_1 G_1(f) + a_2 G_2(f)$$

This means,

$$\mathfrak{F}[a_1 g_1(t) + a_2 g_2(t)] = a_1 G_1(f) + a_2 G_2(f)$$

Linearity of Fourier Transform

More general case

$$\sum_k a_k g_k(t) \Leftrightarrow \sum_k a_k G_k(f)$$

Linearity of Fourier Transform

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$$\sum_k a_k g_k(t) \Leftrightarrow \sum_k a_k G_k(f)$$

Superposition Theorem

Area Under Curve of Fourier Transform $G(f)$

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df$$

Area Under Curve of Fourier Transform $G(f)$

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df$$

Assume $t = 0$

$$g(0) = \int_{-\infty}^{\infty} G(f) e^{j2\pi f \cdot 0} df = \int_{-\infty}^{\infty} G(f) df$$

Area Under Curve of Time domain Function $g(t)$

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-j2\pi ft} dt$$

Area Under Curve of Time domain Function $g(t)$

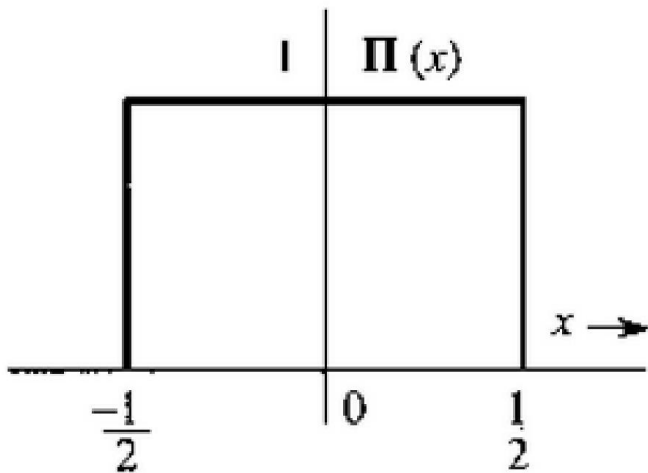
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Fourier Transform of Some Useful Functions

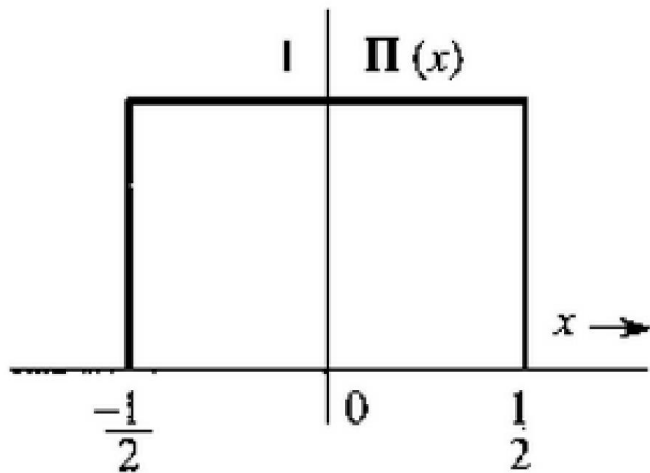
Unit Rectangular Function



$$\Pi(x) = \begin{cases} 1 & |x| \leq \frac{1}{2} \\ 0.5 & |x| = \frac{1}{2} \\ 0 & |x| > \frac{1}{2} \end{cases}$$

Fourier Transform of Some Useful Functions

Unit Rectangular Function

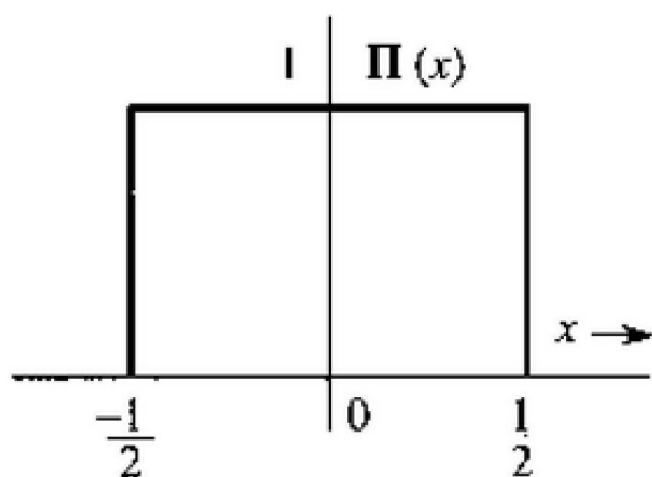


Other names
Box Function
Gate Function

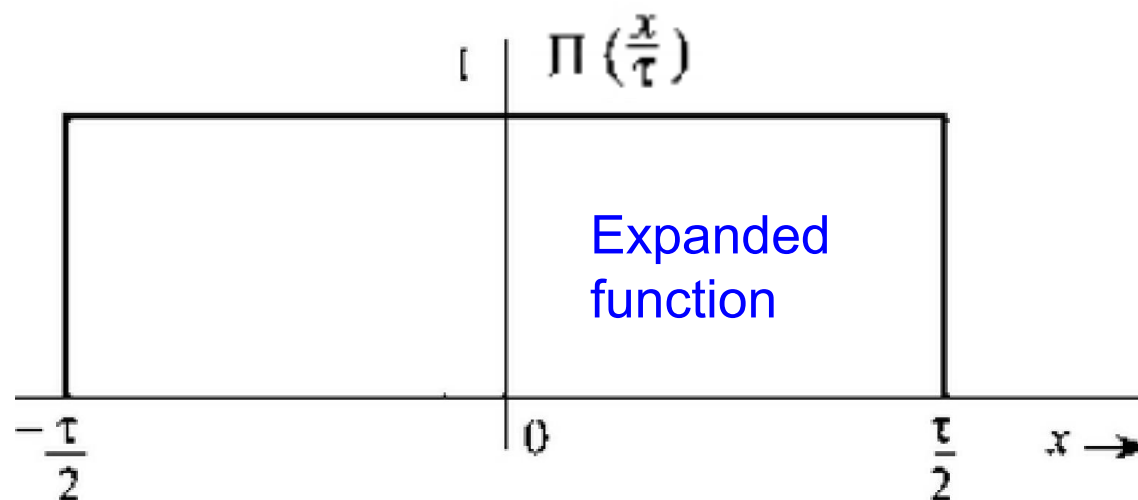
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Fourier Transform of Some Useful Functions

Unit Rectangular Function



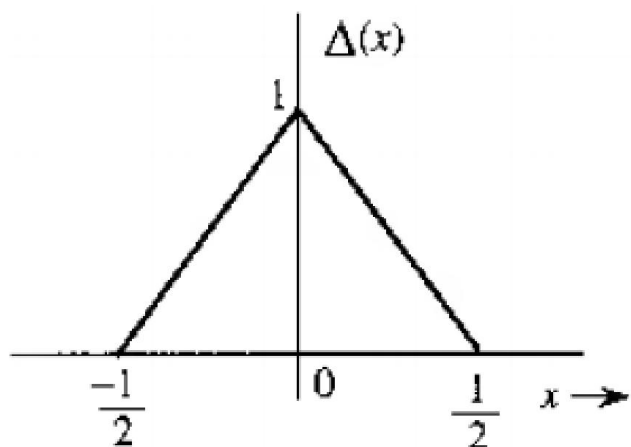
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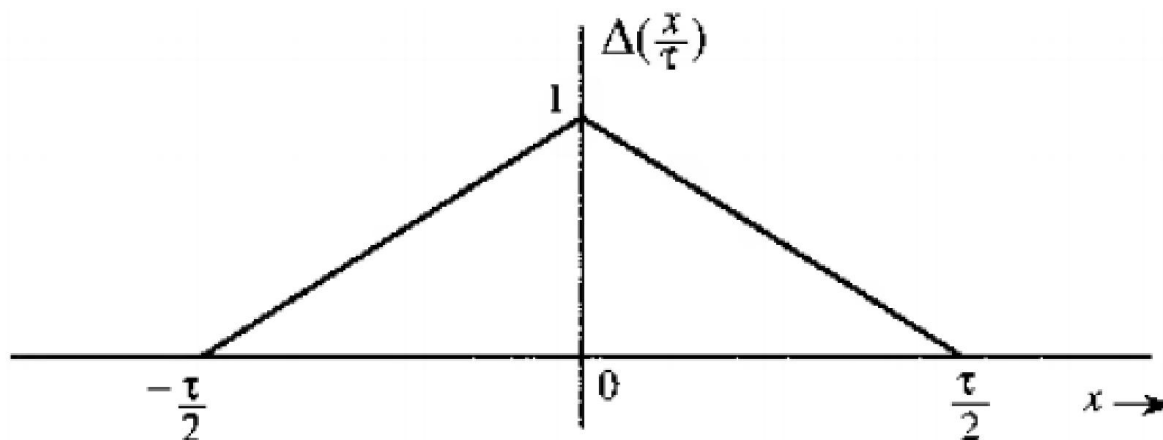
$$\Pi\left(\frac{x}{\tau}\right) = \begin{cases} 1 & |x| \leq \frac{\tau}{2} \\ 0.5 & |x| = \frac{\tau}{2} \\ 0 & |x| > \frac{\tau}{2} \end{cases}$$

Fourier Transform of Some Useful Functions

Unit Triangular Function



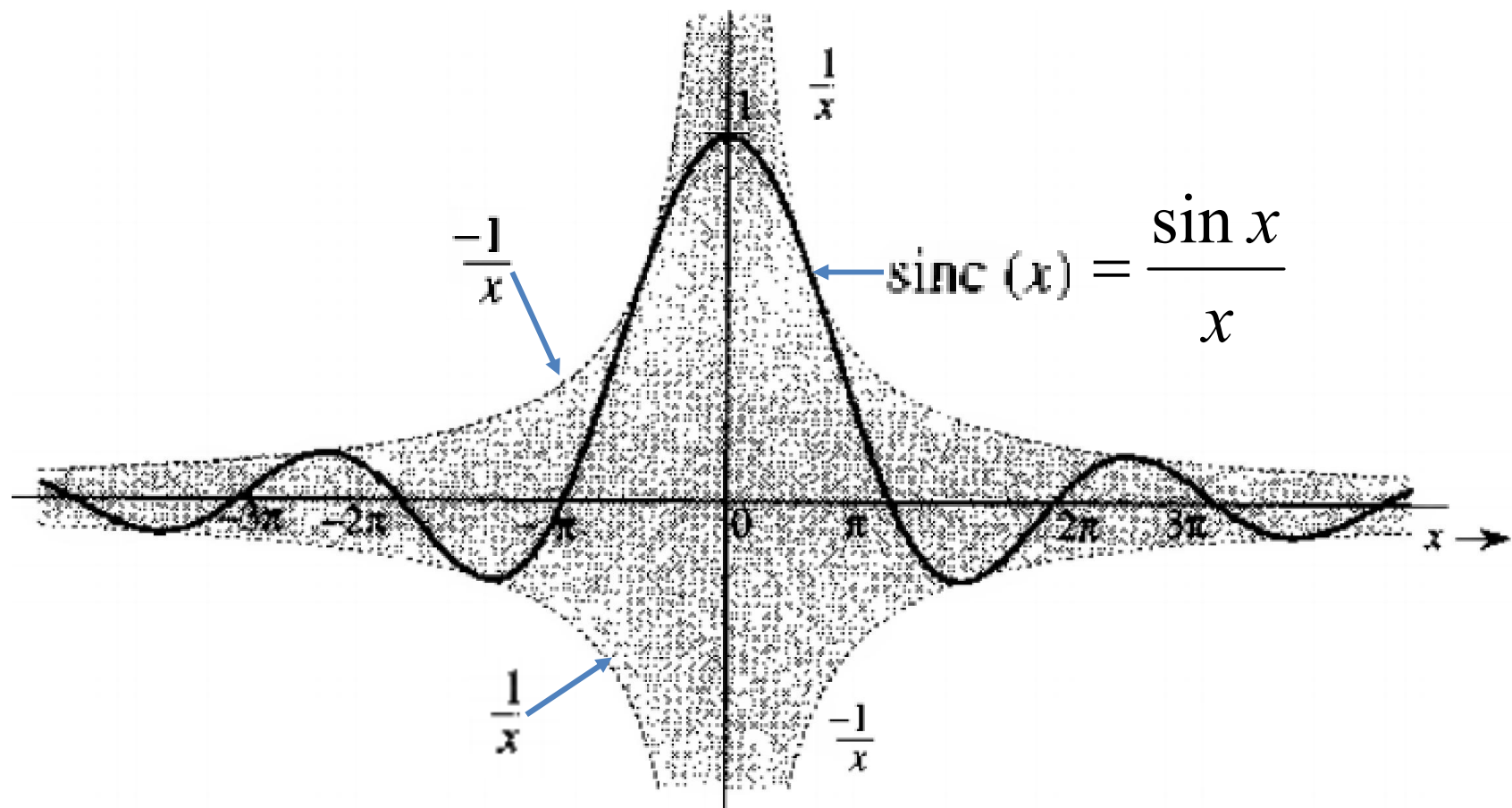
$$\Delta(x) = \begin{cases} 1 - 2|x|, & |x| < \frac{1}{2} \\ 0, & |x| > \frac{1}{2} \end{cases}$$



$$\Delta(\frac{x}{\tau}) = \begin{cases} 1 - 2|\frac{x}{\tau}|, & |x| < \frac{\tau}{2} \\ 0, & |x| > \frac{\tau}{2} \end{cases}$$

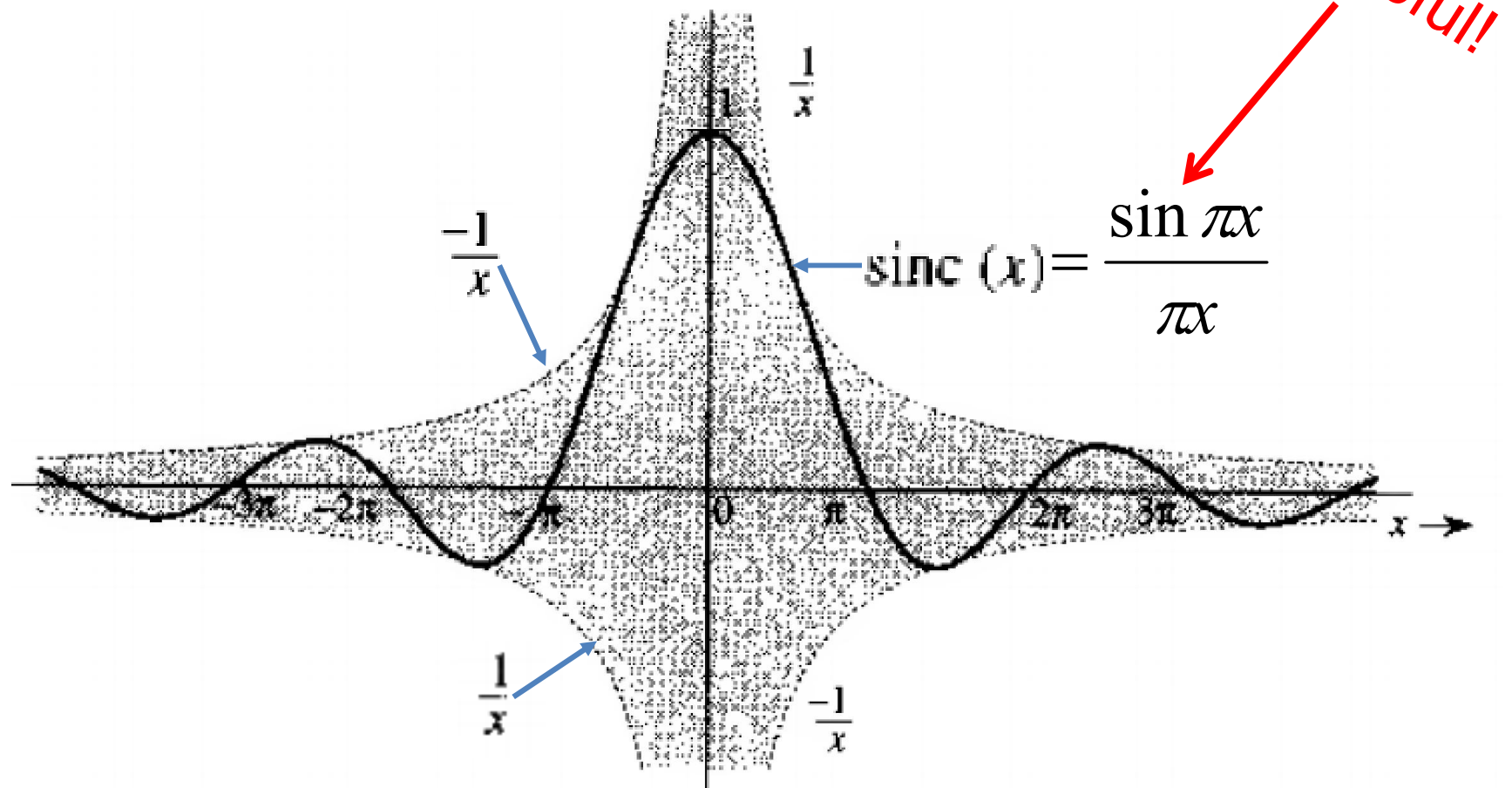
Fourier Transform of Some Useful Functions

Sinc Function, $\text{sinc}(x)$



Fourier Transform of Some Useful Functions

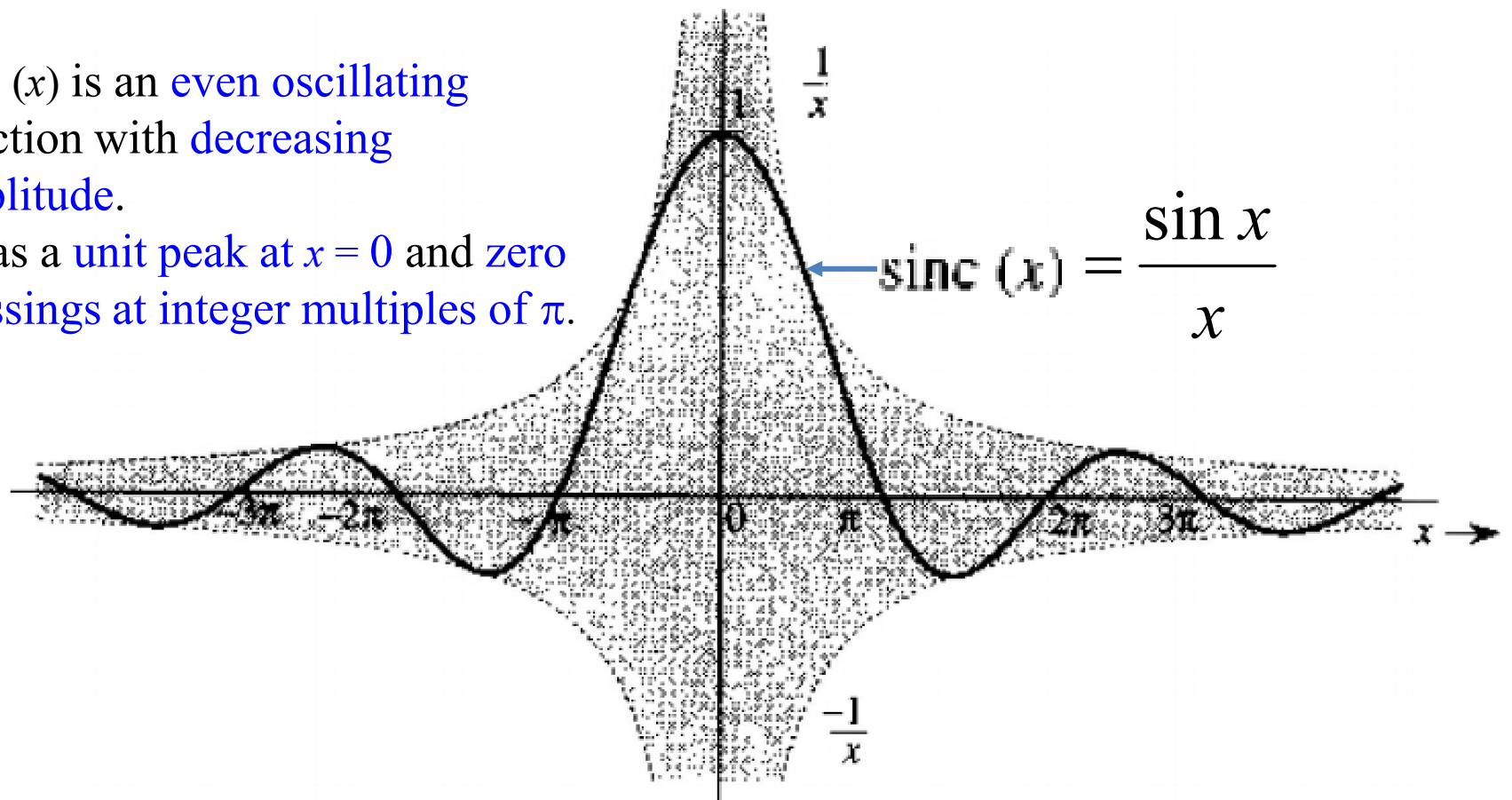
Sinc Function, $\text{sinc}(x)$



Fourier Transform of Some Useful Functions

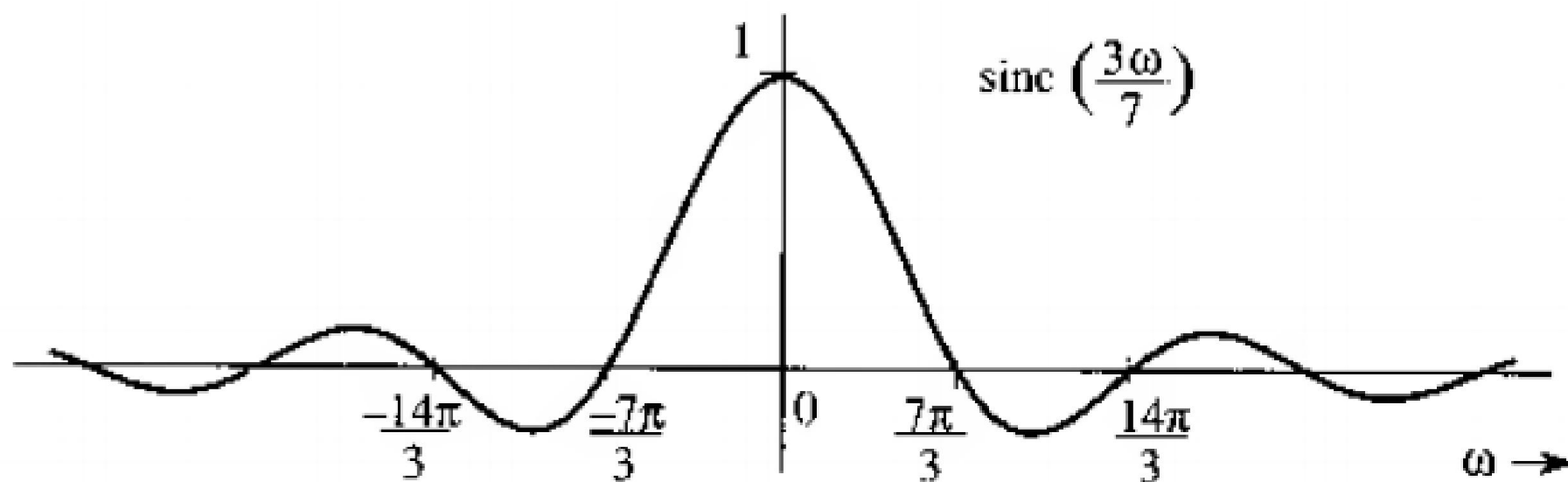
Sinc Function, $\text{sinc}(x)$

- $\text{sinc}(x)$ is an even oscillating function with decreasing amplitude.
- It has a unit peak at $x = 0$ and zero crossings at integer multiples of π .



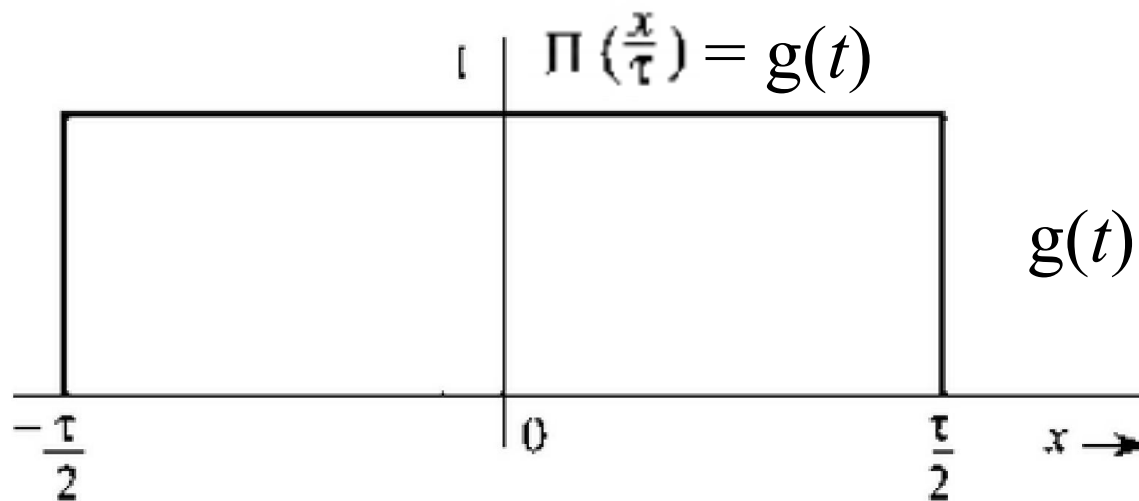
Fourier Transform of Some Useful Functions

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Fourier Transform of Some Useful Functions

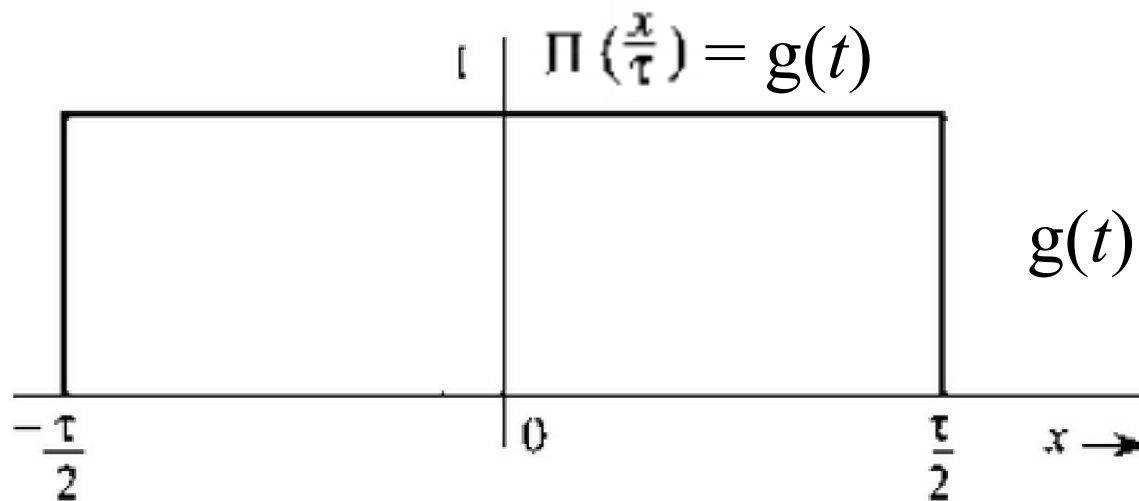
Unit scaled rectangular function



$$g(t) = \Pi\left(\frac{x}{\tau}\right) = \begin{cases} 1 & |x| \leq \frac{\tau}{2} \\ 0.5 & |x| = \frac{\tau}{2} \\ 0 & |x| > \frac{\tau}{2} \end{cases}$$

Fourier Transform of Some Useful Functions

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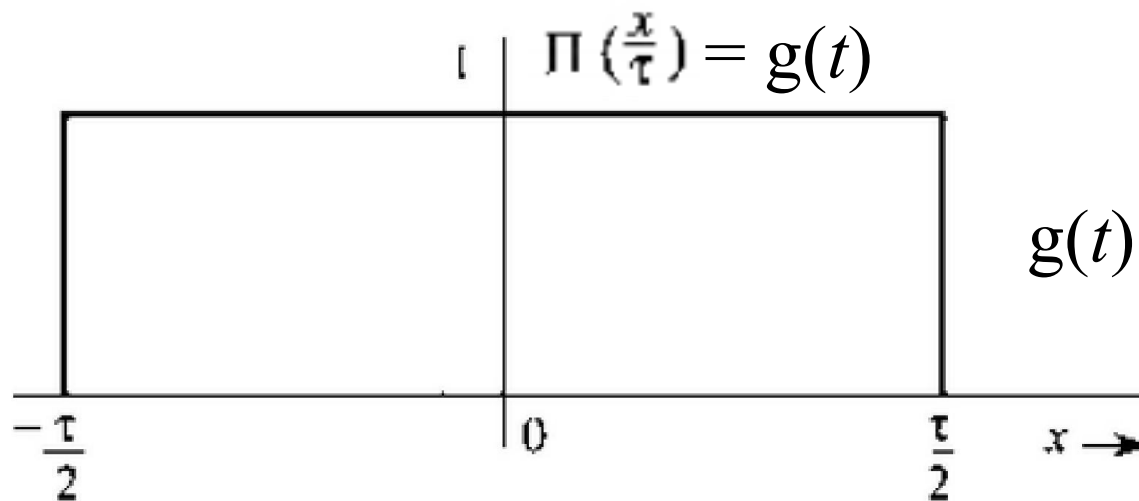


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$$G(f) = \int_{-\infty}^{\infty} \Pi\left(\frac{t}{\tau}\right) e^{-j2\pi ft} dt$$

Fourier Transform of Some Useful Functions

Unit scaled rectangular function

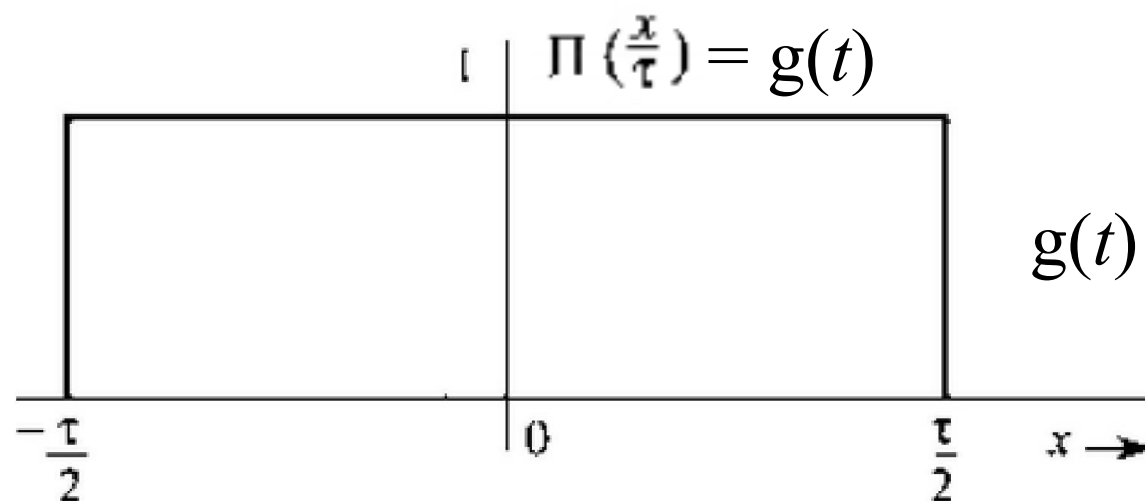


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Fourier Transform of Some Useful Functions

Unit scaled rectangular function

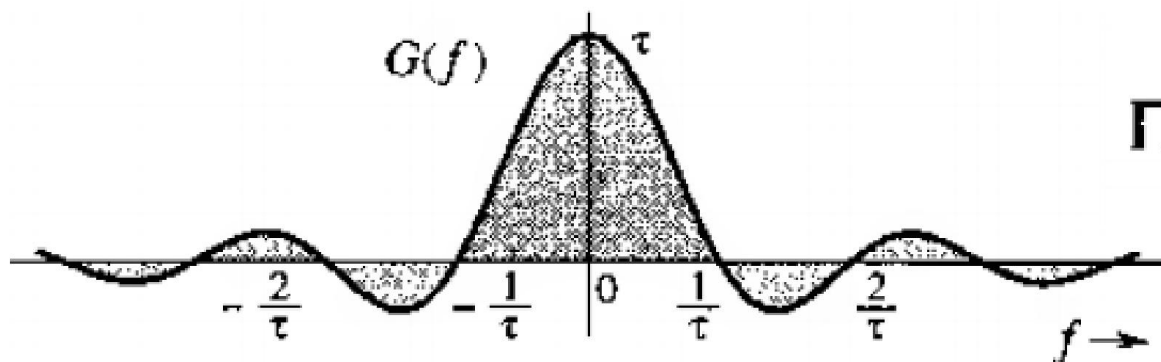
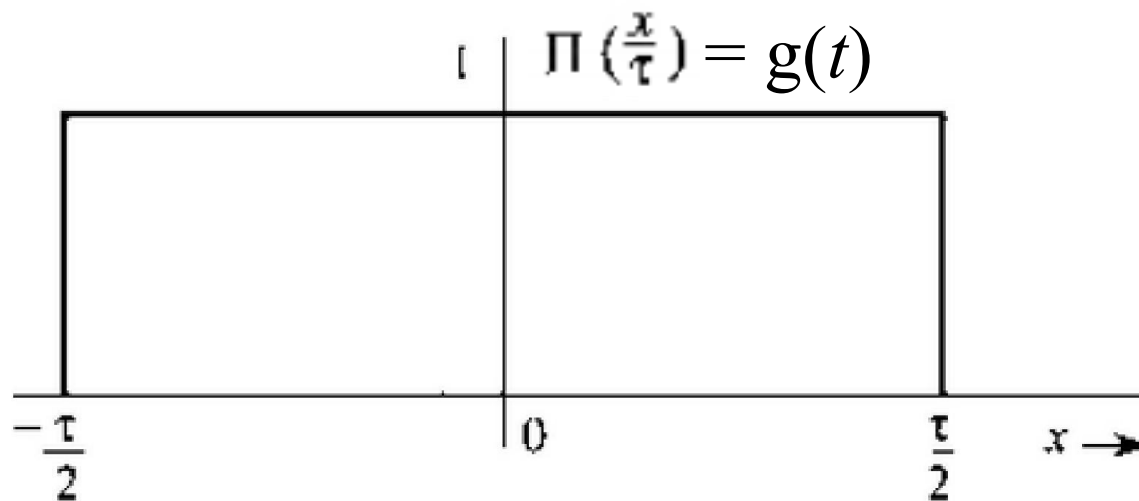


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$$\begin{aligned} G(f) &= \int_{-\infty}^{\infty} \Pi\left(\frac{t}{\tau}\right) e^{-j2\pi ft} dt = \int_{-\tau/2}^{\tau/2} e^{-j2\pi ft} dt \\ &= -\frac{1}{j2\pi f} (e^{-j\pi f \tau} - e^{j\pi f \tau}) = \frac{2 \sin(\pi f \tau)}{2\pi f} = \tau \frac{\sin(\pi f \tau)}{(\pi f \tau)} = \tau \operatorname{sinc}(\pi f \tau) \end{aligned}$$

Fourier Transform of Some Useful Functions

Unit scaled rectangular function

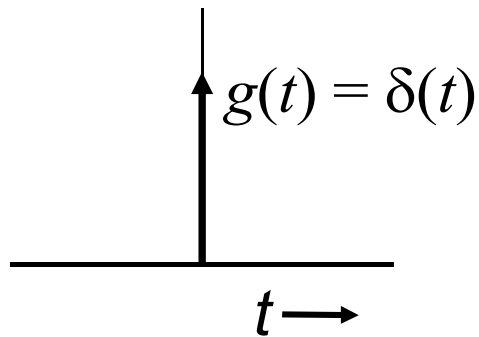


Fourier transform pair

$$\Pi\left(\frac{t}{\tau}\right) \Longleftrightarrow \tau \operatorname{sinc}(\pi f \tau)$$

Fourier Transform of Some Useful Functions

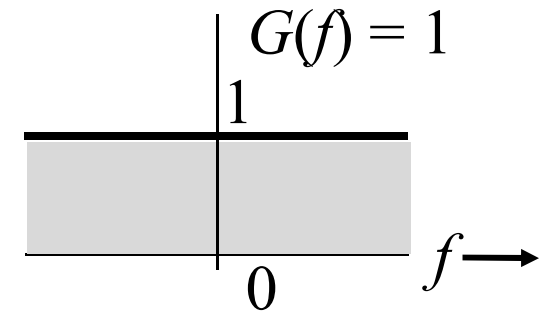
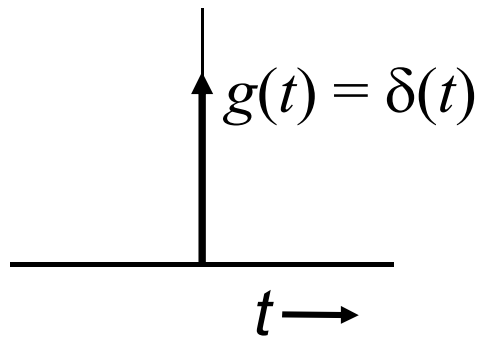
Unit Impulse function $\delta(t)$



$$\begin{aligned}\mathcal{F}[\delta(t)] &= \int_{-\infty}^{\infty} \delta(t) e^{-j2\pi ft} dt \\ &= e^{-j2\pi f \cdot 0} = 1\end{aligned}$$

Fourier Transform of Some Useful Functions

Unit Impulse function $\delta(t)$



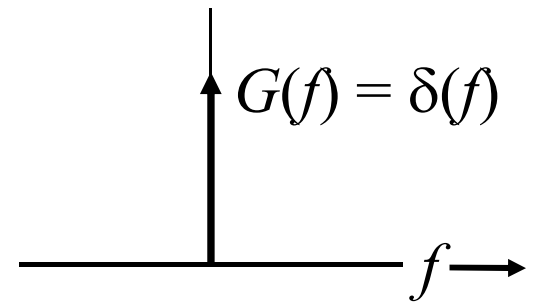
$$\begin{aligned}\mathcal{F}[\delta(t)] &= \int_{-\infty}^{\infty} \delta(t) e^{-j2\pi f t} dt \\ &= e^{-j2\pi f \cdot 0} = 1\end{aligned}$$

Fourier transform pair

$$\delta(t) \longleftrightarrow 1$$

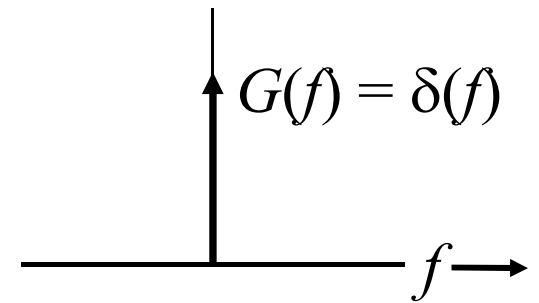
Fourier Transform of Some Useful Functions

Inverse Fourier Transform of $\delta(f)$



Fourier Transform of Some Useful Functions

Inverse Fourier Transform of $\delta(f)$



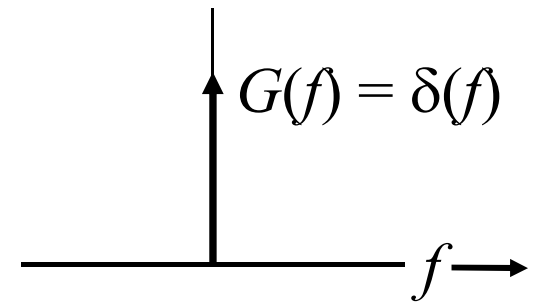
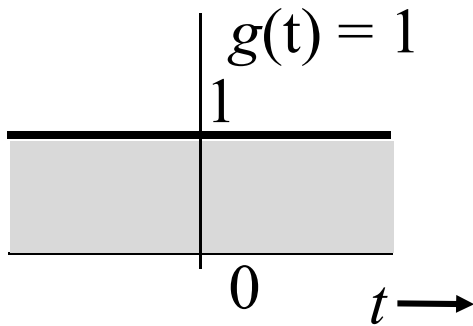
$$g(t) = \mathfrak{F}^{-1}[\delta(f)]$$

$$= \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df = \int_{-\infty}^{\infty} \delta(f) e^{j2\pi ft} df$$

$$= e^{j2\pi \cdot 0} = 1$$

Fourier Transform of Some Useful Functions

Inverse Fourier Transform of $\delta(f)$



$$g(t) = \mathfrak{F}^{-1}[\delta(f)]$$

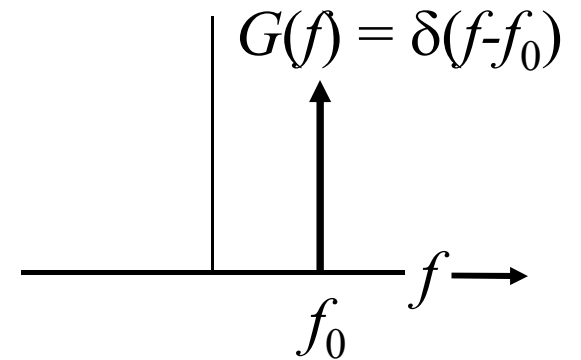
$$\begin{aligned} &= \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df = \int_{-\infty}^{\infty} \delta(f) e^{j2\pi ft} df \\ &= e^{j2\pi \cdot 0} = 1 \end{aligned}$$

Fourier Transform Pair

$$1 \iff \delta(f)$$

Fourier Transform of Some Useful Functions

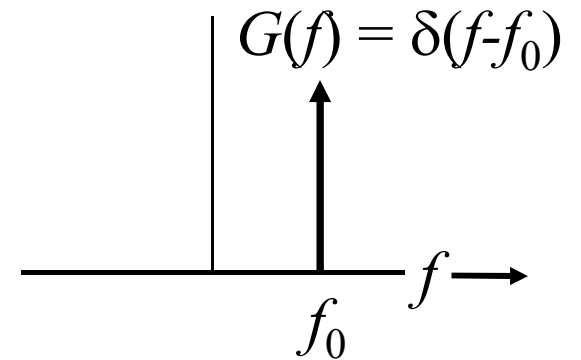
Inverse Fourier Transform of $\delta(f-f_0)$



Fourier Transform of Some Useful Functions

Inverse Fourier Transform of $\delta(f-f_0)$

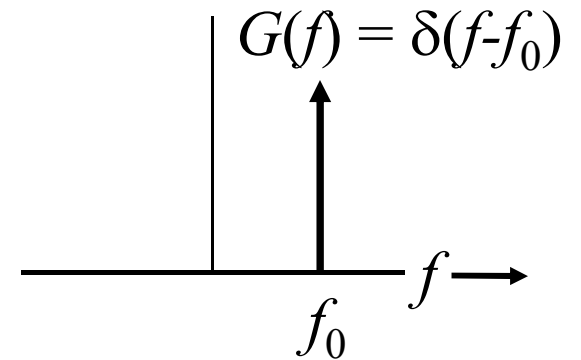
$$\begin{aligned} g(t) &= \mathfrak{F}^{-1}[\delta(f-f_0)] \\ &= \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df = \int_{-\infty}^{\infty} \delta(f-f_0) e^{j2\pi ft} df \\ &= e^{j2\pi f_0 t} = e^{j2\pi f_0 t} \end{aligned}$$



Fourier Transform of Some Useful Functions

Inverse Fourier Transform of $\delta(f-f_0)$

$$\begin{aligned} g(t) &= \mathfrak{F}^{-1}[\delta(f-f_0)] \\ &= \int_{-\infty}^{\infty} G(f) e^{j2\pi f t} df = \int_{-\infty}^{\infty} \delta(f-f_0) e^{j2\pi f t} df \\ &= e^{j2\pi f_0 t} = e^{j2\pi f_0 t} \end{aligned}$$



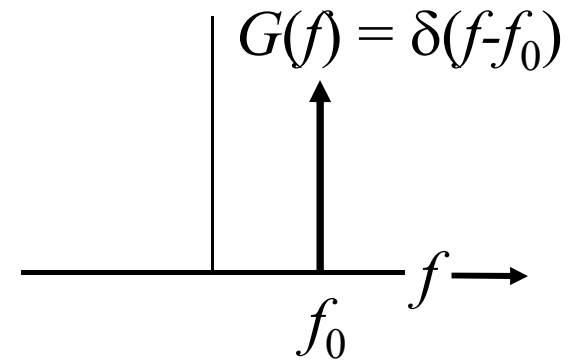
Fourier Transform Pair

$$e^{j2\pi f_0 t} \Leftrightarrow \delta(f-f_0)$$

Fourier Transform of Some Useful Functions

Inverse Fourier Transform of $\delta(f-f_0)$

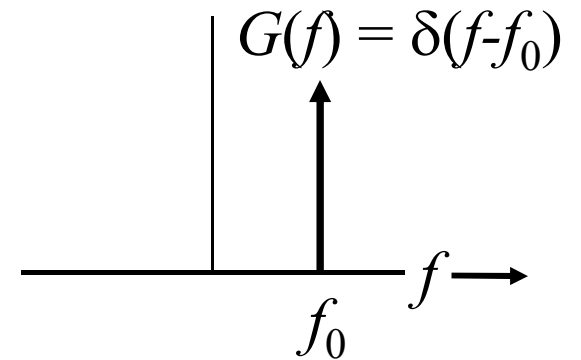
$$e^{j2\pi f_0 t} \Leftrightarrow \delta(f - f_0)$$



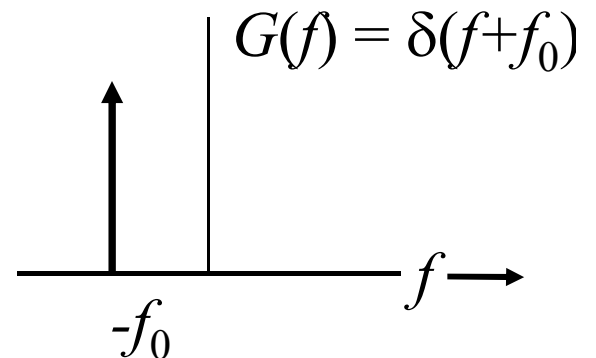
Fourier Transform of Some Useful Functions

Inverse Fourier Transform of $\delta(f-f_0)$ and $\delta(f+f_0)$

$$e^{j2\pi f_0 t} \Leftrightarrow \delta(f - f_0)$$



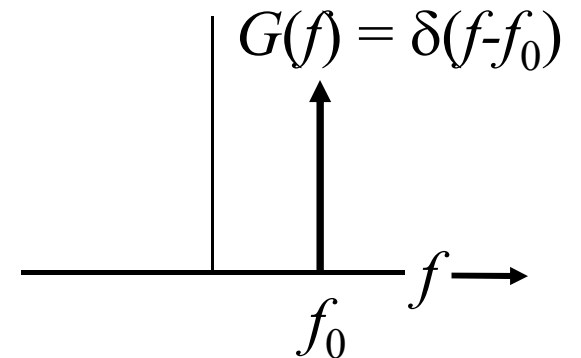
$$? \Leftrightarrow \delta(f + f_0)$$



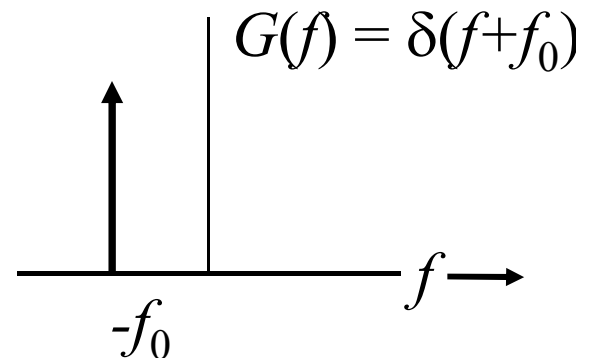
Fourier Transform of Some Useful Functions

Inverse Fourier Transform of $\delta(f-f_0)$ and $\delta(f+f_0)$

$$e^{j2\pi f_0 t} \Leftrightarrow \delta(f - f_0)$$



$$e^{-j2\pi f_0 t} \Leftrightarrow \delta(f + f_0)$$



Fourier Transform Pairs

Fourier Transform of Some Useful Functions

Fourier Transform of cosine function

Assuming, $e^{j\theta} = \cos \theta + j \sin \theta$

$$\cos 2\pi f_0 t = \frac{1}{2} (e^{j2\pi f_0 t} + e^{-j2\pi f_0 t})$$

Fourier Transform of Some Useful Functions

Fourier Transform of cosine function

Assuming, $e^{j\theta} = \cos \theta + j \sin \theta$

$$\cos 2\pi f_0 t = \frac{1}{2} (e^{j2\pi f_0 t} + e^{-j2\pi f_0 t})$$

Fourier Transform Pair

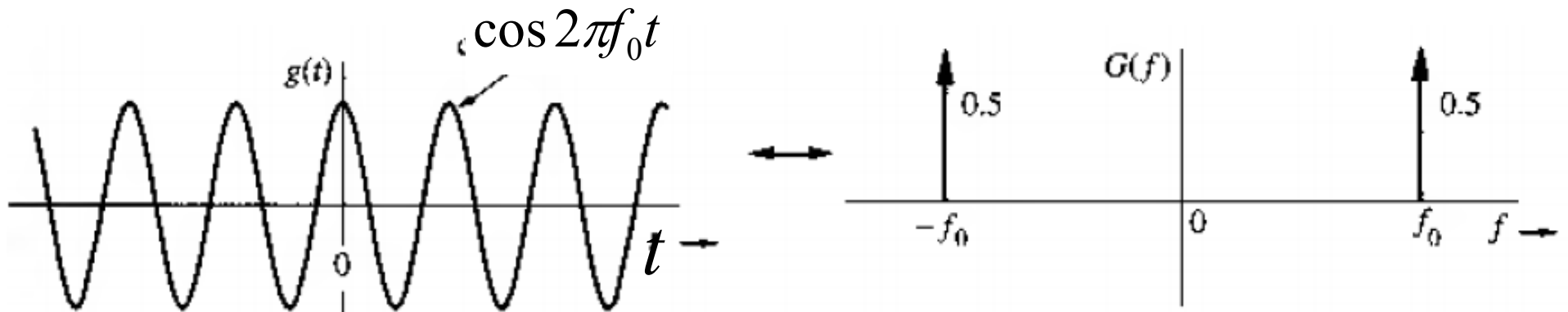
$$\cos 2\pi f_0 t \Leftrightarrow \frac{1}{2} [\delta(f - f_0) + \delta(f + f_0)]$$

Fourier Transform of Some Useful Functions

Fourier Transform of cosine function

Assuming, $e^{j\theta} = \cos \theta + j \sin \theta$

$$\cos 2\pi f_0 t = \frac{1}{2} (e^{j2\pi f_0 t} + e^{-j2\pi f_0 t})$$



$$\cos 2\pi f_0 t \Leftrightarrow \frac{1}{2} [\delta(f - f_0) + \delta(f + f_0)]$$

Fourier Transform of Some Useful Functions

Fourier Transform of sine function

Assuming, $e^{j\theta} = \cos \theta + j \sin \theta$

$$\sin 2\pi f_0 t = \frac{1}{2j} (e^{j2\pi f_0 t} - e^{-j2\pi f_0 t})$$

Fourier Transform of Some Useful Functions

Fourier Transform of sine function

Assuming, $e^{j\theta} = \cos \theta + j \sin \theta$

$$\sin 2\pi f_0 t = \frac{1}{2j} (e^{j2\pi f_0 t} - e^{-j2\pi f_0 t})$$

Fourier Transform Pair

$$\sin 2\pi f_0 t \Leftrightarrow \frac{1}{2j} [\delta(f - f_0) - \delta(f + f_0)]$$